

Lecture 2 (9/16)

Independence, Marginalization, and Conditioning

Example

$P(C, G)$	$G = M$	$G = F$
$C = 1$ (diagnosed)	0.0021	0.0019
$C = 0$ (not diagnosed)	0.4899	0.5061

For example, if these numbers are obtained from a population of N people,

$$0.0021 = \frac{N(\{\text{diagnosed \& Male}\})}{N}$$

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What we care about depends on who we are:

- Hospital management: What is the chance a random person (M or F) is diagnosed? (marginalization)
- Dr: What is the chance my patient (known to be M or F) will be diagnosed? (conditional probability)

For the hospital,

$$\begin{aligned}P(C = 1) &= P(C = 1, G = M) + P(C = 1, G = F) \\&= 0.0021 + 0.0019 = 0.004\end{aligned}$$

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We used that probability of disjoint events as: **(Countable) additivity**: for any countable collection (meaning we can list them as $E_1, E_2, \dots$) of pairwise disjoint sets $E_1, E_2, \dots$,

$$P\left(\bigcup_i E_i\right) = \sum_i P(E_i).$$

Marginalization and independence

Sum out the variable you don't care about:

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Cancer diagnosis data: compare the joint probabilities to the products of the marginals:

(C, G)	Joint $P(C, G)$	Product $P(C)P(G)$
$(1, M)$	0.0021	0.001968
$(1, F)$	0.0019	0.002032
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$(0, F)$	0.5061	0.505968

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- One way to approach: collect more data and see if the numbers get closer
- If we can't do this, we need to quantify statistical uncertainty (Unit 3)

Example

$X, Y \in \{0, 1\}$ have joint distribution

$$P(X = x, Y = y) = \begin{cases} 1/2 & (0, 0) \\ 1/8 & (0, 1) \\ 1/8 & (1, 0) \\ 1/4 & (1, 1) \end{cases}$$

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Example

Now suppose $X \in \{0, 1\}$ and $Y \in \{1, 2, 3, 4\}$ have joint distribution

$P(X, Y)$	$Y = 1$	$Y = 2$	$Y = 3$	$Y = 4$
$X = 0$	$1/4$	$1/8$	$1/8$	0
$X = 1$	0	$1/8$	$1/8$	$1/4$

Are X and Y independent?

Conditioning

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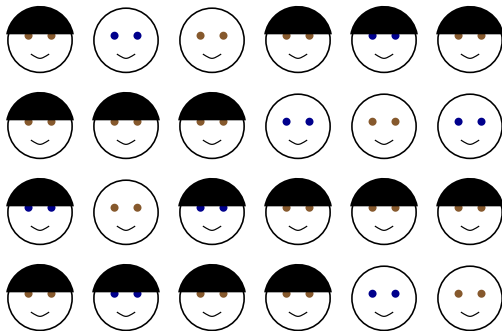
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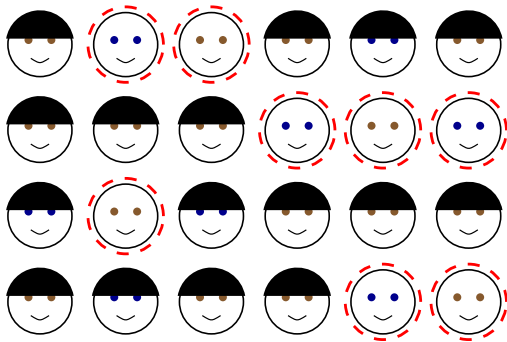
$P(x | y)$ is the fraction of times x is observed among outcomes for which y is observed.

- Marginalize over X means: I no longer see X
- Conditioning on X : I fix X to a certain value

Marginalizing vs. conditioning: a population



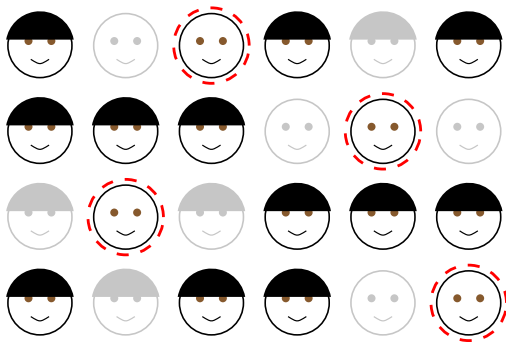
Marginalizing vs. conditioning: a population



Marginalizing over eye color.

$$P(\text{bald}) = \frac{8}{24} = \frac{1}{3}.$$

Marginalizing vs. conditioning: a population



Conditioning on brown eyes.

$$P(\text{bald} \mid \text{brown}) = \frac{4}{16} = \frac{1}{4}.$$

Example

Cancer diagnosis data again

$C \mid G$	Marginal $P(C)$	Conditional $P(C \mid G)$
1 M	0.0040	0.00427
1 F	0.0040	0.00374
0 M	0.9960	0.9957
0 F	0.9960	0.9963

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- Are they independent? Same problem as marginal table.
- But it is much easier to see the dependence:

$$\text{effect of being male} = P(1|M) - P(1|F) = 0.00427 - 0.00374 = 0.00053$$

Taste of regression models

- Regression models¹ are essentially models given to us as conditional probability models.

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- Thus it is natural to consider models given to us in conditional form: For a pair of Bernoullis (X, Y) , instead of defining a joint distribution, we can write

$$\begin{aligned} X &\sim \text{Bernoulli}(q_X) \\ Y|X &\sim \text{Bernoulli}(q_1 X + q_0(1 - X)) \quad q_1, q_0 \in [0, 1] \end{aligned}$$

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- $Y|X$ means the distribution of Y conditioned on a value of X
- In general

$$Y|X \sim \text{Name}(\text{Parameters})$$

where now “Parameters” depends on X (if they are not independent).

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Example

From the cancer diagnosis data we saw (coding $G = 1$ for male, $G = 0$ for female)

$$q_1 = P(C = 1 \mid G = 1) = 0.00427, \quad q_0 = P(C = 1 \mid G = 0) = 0.00374.$$

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Example

Consider

$$P(X = x, Y = y) = \begin{cases} 1/2 & (0, 0) \\ 1/8 & (0, 1) \\ 1/8 & (1, 0) \\ 1/4 & (1, 1) \end{cases}$$

Write a conditional model for $Y|X$.

Example

$$X \sim \text{Bernoulli}(1/3)$$
$$Y|X \sim \text{Bernoulli}(X/2 + 1/3)$$

Find all the joint probabilities and the marginal of Y .

Example

We are betting on a series of three throws, each modeled as independent Bernoulli variable $X_i \sim \text{Bernoulli}(q)$.

- What is the chance we see a run of exactly 4?
- What is the chance we get 4 more after seeing a run of 6?

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- What is the chance we see a run of exactly 4?
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$$P(\text{run of 4}) = P(X_1 = X_2 = X_3 = X_4 = 1, X_5 = 0) = q^4(1 - q)$$

$$P(\text{run of 4 more} | \text{run of 6}) = \frac{P(X_1 = \dots = X_{10} = 1, X_{11} = 0)}{P(X_1 = \dots = X_6 = 1)} = \frac{q^{10}(1 - q)}{q^6}$$

A Markov Chain

Example

$$X_0 \sim \text{Bernoulli}(1/2)$$

$$X_{t+1} \mid (X_t, X_{t-1}, \dots, X_0) \sim \text{Bernoulli}(q_1 X_t + q_0(1 - X_t))$$

and assume $X_0 = 1$.

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and assume $X_0 = 1$. Let $p_t = P(X_t = 1)$. Then

$$\begin{aligned} p_t &= P(X_t = 1 \mid X_{t-1} = 1)P(X_{t-1} = 1) + P(X_t = 1 \mid X_{t-1} = 0)P(X_{t-1} = 0) \\ &= q_1 p_{t-1} + q_0(1 - p_{t-1}) = (q_1 - q_0)p_{t-1} + q_0. \end{aligned}$$

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Hence, the probabilities no longer change when $p_t = p_{t-1}$,

$$p_t = \frac{q_0}{1 - q_1 + q_0}.$$

Bayes' formula

Bayes' formula relates the two conditional probabilities for a pair of random variables:

$$P(x | y) = \frac{P(y | x) P(x)}{P(y)}.$$

Example

A disease affects 1% of a population. A test is positive 95% of the time if you have the disease, and 10% of the time if you don't. Find $P(D \mid +)$.

Example

It rains on 20% of days. When it rains, the forecast correctly predicts rain 90% of the time. When it doesn't rain, the forecast incorrectly predicts rain 15% of the time. Find $P(\text{rain} \mid \text{forecast rain})$.